Context of Business:

2Market, a global supermarket (online & instore), seeks to understand customer demographics, purchasing preferences, and advertising performance. This will help allocate the advertising budget effectively, targeting the right customer base to increase profits.

A. The Data:

We have two datasets: marketing_data having customer demographics, purchases, and marketing feedback, and ad_data with advertising effectiveness.

B. Further Data for Improved Analysis:

1. Analyzing specific products and brands will help identify top-sellers and inform future strategies.
2. Splitting lead conversions between marketing and advertising offers better insights.
3. Assessing the cost of marketing and advertising campaigns ensures profitability analysis as well.

Analytical Approach:

Excel and SQL were extensively used for data preprocessing and manipulation to aid analytics development.

A. Data Manipulation:

1. An **Age** variable was created by subtracting the Year_Birth column from the current year. This was bucketed into:
    - 27–45
    - 46–65
    - 66+

```sql
/* Creating Age variable */

ALTER TABLE marketing_data
ADD "Age" INTEGER;

UPDATE marketing_data
SET "Age" = EXTRACT(YEAR FROM CURRENT_DATE) - "Yea
|
Select * from marketing_data;

/* Creating Age Bucket variable */

-- Observing the Age Distribution--
SELECT "Age", count("Age")
FROM marketing_data
GROUP BY "Age"
ORDER BY "Age" ASC

--Creating Buckets--

ALTER TABLE marketing_data
ADD COLUMN "Age Bucket" VARCHAR(15);

UPDATE marketing_data
SET "Age Bucket" = CASE
    WHEN "Age" BETWEEN 27 AND 45 THEN '27-45'
    WHEN "Age" BETWEEN 46 AND 65 THEN '45-65'
    ELSE '66 +'
END;
```

2. The **Income** variable was analyzed and bucketed based on the 37th and 66th percentiles:
    - Low Income (0–37th percentile)
    - Middle Income (37th–66th percentile)

- High Income (66th+ percentile)

```sql
/* Creating Income Bucket variable */

-- Observing the Income Distribution--
SELECT Max("Income") as "Max Income", min("Income") as "Min Income"
FROM marketing_data

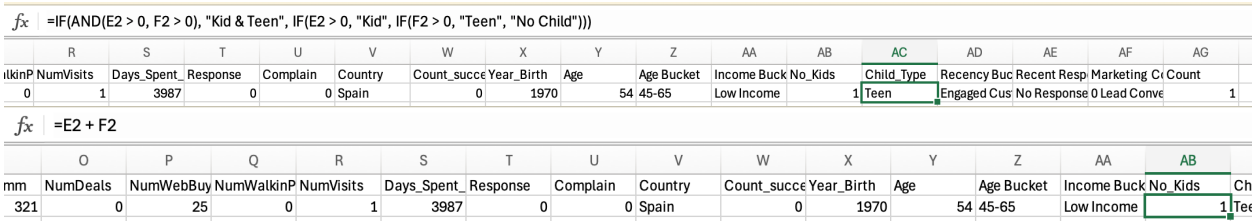
--Creating Buckets--
WITH Income_Perc AS (
    SELECT
        PERCENTILE_CONT(0.37) WITHIN GROUP (ORDER BY "Income") AS Perc_37,
        PERCENTILE_CONT(0.66) WITHIN GROUP (ORDER BY "Income") AS Perc_66
    FROM marketing_data
)

-- Creating Buckets based on Percentiles
ALTER TABLE marketing_data
ADD COLUMN "Income_Bucket" VARCHAR(15);

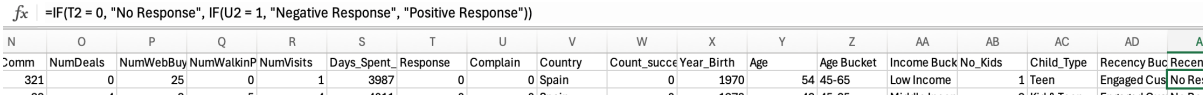
-- Updating the Income Bucket based on calculated percentiles
UPDATE marketing_data
SET "Income_Bucket" = CASE
    WHEN "Income" <= (SELECT Perc_37 FROM Income_Perc) THEN 'Low Income'
    WHEN "Income" <= (SELECT Perc_66 FROM Income_Perc) THEN 'Middle Income'
    ELSE 'High Income'
END;
```

3. A new **Child_Type** variable was created using Kidhome and Teenhome, with categories:
    - Kid (Households with children)
    - Teen (Households with teenagers)
    - Kid & Teen (Both children and teenagers)
    - No Child (No children or teenagers)
    -

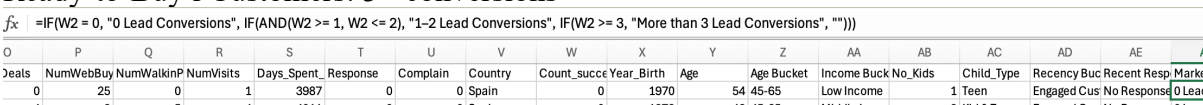
Additionally, a **No_Kids** variable was created to count total kids/teens in each household.



4. A **Recent Response** variable was created using the Response and Complain columns:
    - "Positive Response" , "Negative Response" (1 values in Response and Complain)
    - "No Converted" (0 values)

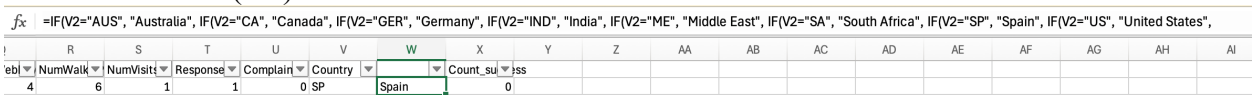


5. A **Marketing Conversion Bucket** was created based on count_success (Total lead conversions)(later created in Tableau as Calculated field):
    - Hard-to-Convert Customers: 0 conversions
    - On-the-Fence Customers: 1–2 conversions
    - Ready-to-Buy l Customers: 3+ conversions

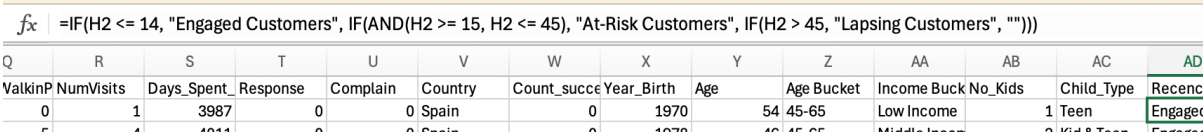


(later created in Tableau as Calculated field with "Hard-to-Convert" etc)

6. A count variable was created (value of 1 for each row) to help with further calculations.
7. The **Country** variable created using the country codes, with the following categories:
    - Australia (AUS)
    - Canada (CA)
    - Germany (GER)
    - India (IND)
    - Middle East (ME)
    - South Africa (SA)
    - Spain (SP)
    - United States (US)



8. The **Recency** variable was analyzed, and the **Recency Bucket** was created:
    - Engaged Customers: 0–14 days
    - At-Risk Customers: 15–45 days
    - Lapsing Customers: 45+ days



B. Data Preprocessing:

Involved removing leading/trailing spaces in marketing_data and ad_data, inserting cleaned data into new tables, identifying duplicates by ID, validating date values, and fetching distinct values from specific columns for further analysis.

```sql
/* Removing Trailing & Leading Spaces */

INSERT INTO PUBLIC.marketing_data_clean (
    "ID"
    ,"Education"
    ,"Marital_Status"
    ,"Country"
    )
SELECT TRIM("ID") AS "ID"
    ,TRIM("Education") AS "Education"
    ,TRIM("Marital_Status") AS "Marital_Status"
    ,TRIM("Country") AS "Country"
FROM marketing_data;

SELECT *
FROM PUBLIC.marketing_data_clean;

INSERT INTO PUBLIC.ad_data_clean ("ID")
SELECT TRIM("ID") AS "ID"
FROM ad_data;

SELECT *
FROM PUBLIC.ad_data_clean;
```

```sql
/* Check for duplicates in PK */

SELECT "ID"
    ,count(*)
FROM ad_data_clean
GROUP BY "ID"
HAVING count(*) > 1;

SELECT "ID"
    ,count(*)
FROM marketing_data_clean
GROUP BY "ID"
HAVING count(*) > 1;
```

```sql
/* Check for correctness in date values */

SELECT DISTINCT "year"
FROM (
    SELECT "Dt_Customer"
        ,extract('year' FROM "Dt_Customer") AS year
    FROM marketing_data
    );

SELECT DISTINCT "month"
FROM (
    SELECT "Dt_Customer"
        ,extract('month' FROM "Dt_Customer") AS month
    FROM marketing_data
    );

SELECT DISTINCT "day"
FROM (
    SELECT "Dt_Customer"
        ,extract('day' FROM "Dt_Customer") AS day
    FROM marketing_data
    );

/* Checking Distinct Values */
```

```sql
/* Checking Distinct Values */

SELECT DISTINCT Education
FROM marketing_data;

SELECT DISTINCT Country
FROM marketing_data;

SELECT DISTINCT Marital_Status
FROM marketing_data;
```

C. Data Integration:

Joined the two tables for marketing and advertising using inner joins.

```sql
/* Joining Marketing & Advertisement Data */

SELECT *
FROM marketing_data_clean
INNER JOIN ad_data_clean USING ("ID");
```

Dashboard Development:

This Tableau dashboard was developed to visually present various data groupings, helping identify insights and stories within the business. The goal was to use such stories to inform decision making and resolve 2 market's business problem.

To achieve this, dataset attributes were clustered based on their functional relevance. The dashboard aggregates data across key areas, including customer demographics, customer engagement, purchase behavior, marketing response, and advertising effectiveness, propensity & purchase recency.

A. Motive, Variables & Layout:

Section 1: Key Metrics – This section offers a concise overview of the quantitative measures that provide insights into customer preferences at 2Market and the effectiveness of advertising and marketing efforts.

Section 2: Customer Demographics – This section provides insights into customer characteristics through demographic statistics (age, education, marital status etc.). It presents grouping of customer counts based on such variables.

Section 3: Customer-Company Interactions with Customer Demographics– Similar to above, it explores how customer-company engagement metrics (such as purchases, visits, advertising & marketing responses etc.) vary across demographic factors.

Section 4: Customer-Company Interactions with Propensity & Purchase Recency – It focuses on relationships between propensity (ready to buy etc. customers), recency (customer status: actively purchasing, lapsing, etc.) across engagement metrics, purchasing behavior, and marketing/advertising attributes.

B. Charts and Visuals:

Bar Charts – Vertical and horizontal bar charts are used in the first two sections to compare category counts clearly.

Bubble Charts – Used in the third section to highlight customer- company engagement variations across propensity & purchase recency. The varying bubble sizes emphasize the significance of each metric.

Donut Charts – Two donut charts were implemented to represent the composition of marketing response within propensity & purchase metrics.

Key Metrics – Interactive large numbers and text were used here.

C. Color Scheme:

Gradient colors highlight value changes, with darker shades indicating higher values. Blue hues are consistently used for customer engagement and interaction metrics, creating a visually cohesive experience.

D. Interactivity:

Filters, dropdown menus and navigation panes are used to make the dashboard interactive by controlling what tiles, graphs and visuals are displayed. This gives users a more GUI-like experience for exploring different sections. Additionally, hover-over infographics in the donut charts provide information as needed, and reducing clutter of information.

E. Accessibility:

The use of filters, labels, tool tips, appropriate color, size and font, makes this dashboard accessible. Care has been taken not overload the dashboard with graphs, ensuring a clear and user-friendly experience.

Patterns, Trends & Insights:

A. High-Engagement Customer Demographics

- **High earning**, **married Spanish** customers aged **45-65** with **graduate** degrees are most engaged with 2 Market and they consume beverages mostly.
- Overall **Beverages** is the most consumed product followed by **Meat** items

B. Marketing & Advertising Influence on Customer Segments

- The highly engaged customer demographics above also responded positively to advertisements and marketing. But we see as slight change in family demographics - **single individuals** and those **without children** respond more to **recent marketing** efforts, likely due to **product relevance** for single-person or childless households.
- Overall **Twitter** is the best advertising channel; however, many **high engaged** customers prefer **Instagram**.
- Avg **website conversions outrun** those by **marketing.**

C. Engagement and Purchasing Behavior by Propensity

- **Lower-propensity customers**, despite not being "ready-to-buy," displayed significant engagement and purchasing behavior, calling into **question** the **effectiveness of prior marketing** or **adverting strategies.** Recent campaigns support this fact once more, with so expressing dissatisfaction.
- Campaign however is moderately successful (~ 50% conv.) on "on-the-fence" customers and a few high -propensity customers declined the offer.

D. Insights from Purchase Recency Patterns

- Previously highly engaged customers are showing signs of disengagement, while less engaged customers are becoming more active.
- **Disengagement** may result from **seasonal buying** behavior or **new product** strategies that attracted different customers but alienated loyal ones- **need** more **analysis**.
- **Lapsing customers,** primarily low-propensity buyers, are not responding well to recent marketing, indicating a **need** for **re-engagement** strategies**.**